DELHI MRTS PROJECT PHASE III LINE 7

TRAIN CONTROL & SIGNALLING SYSTEM CONTRACT CS03

TRACK AND SIGNALLING PLAN LINE 7 (MUKUNDPUR DEPOT, SECTION - MD)

DELHI METRO RAIL CORPORATION LTD.
Metro Bhawan, Fire Brigade Lane, Barakhamba Road, New Delhi-110001

ORIGINATOR:
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AS-BUILT DRAWING

REV	DATE	DESCRIPTION
6.0	01-APR-20	AS-BUILT DRAWING

	NAME	DATE
DESIGNED	KULDEEP M.	01-APR-20
CHECKED	ASHOK K.	01-APR-20
APPROVED	ROHIT T.	01-APR-20

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TRAIN CONTROL & SIGNALLING SYSTEM CONTRACT CS03
TRACK AND SIGNALLING PLAN LINE 7
MUKUNDPUR DEPOT

DRAWING No:	COST CENTRE:	SCALE	SHEET	CONT.	REV.
100235879	A PRELIMINARIES AND GENERAL REQUIREMENTS	N/S	1	2	6.0

REV.	REVISION INFORMATION	DESIGNED	CHECKED	DATE
6.0	AS-BUILT RELEASED	KULDEEP S.	ASHOK K.	01-APR-2020
5.3	UPDATED AS PER CBTC MAP SITE SURVEY VERIFICATION REPORT_R8.DOCX VER 6.0.	KULDEEP S.	ASHOK K.	01-APR-2020
	UPDATED AS PER CBTC MAP SITE SURVEY VERIFICATION REPORT_R9.DOCX VER 2.0.			
5.2	UPDATED AS PER CBTC MAP SITE SURVEY VERIFICATION REPORT_R8.DOCX VER 3.0	KULDEEP S.	JAYA S	27-JUN-2017
5.1	MOVED P0901 FROM LHS TO RHS AS PER NCR#203	JAYA S.	VARUN C.	08-MAR-2017
	REMOVED AE & AB MARKERS AT ETU BUILDING. REMOVED NORMING POINT FROM AX SECTIONS T08127 & T08129			
	(REFER FCO-DL7-2017-013, NCR: bkk_067#686)			
	UPDATED PSEUDO PLATFORM PP08825 AT AUTO COACH WASH PLANT			
	ADDED SPK -35 & SPK-34 AT HEAVY INTERNAL CLEANING BUILDING & PIT WHEEL LATHE RESPECTIVELY(NCR#518			
	AND FC0-DL7-2017-012)			
5.0	RELEASED	JAYA S.	ASHOK K.	05-SEP-2016
4.6	ADDED WNRA08712,WNRA08713,WNRA08714 & WNRA08715 AS PER RF SCHEMATIC REV 6.0	JAYA S.	ASHOK K.	06-MAY-2016
	ADDED CCTV08717,CCTV08718,CCTV08719 & CCTV08720 AS PER RF SCHEMATIC REV 6.0			
	UPDATED AS PER NCR#bkk_067_400			

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REV.	REVISION INFORMATION	DESIGNED	CHECKED	DATE
4.5	ADDED CCTV08015 AND CCTV08016 AS PER RF SCHEMATIC REV 5.3	JAYA S.	VARUN C.	11-JAN-2016
4.4	UPDATED THE BLOCK OF "POINT"- POINT MACHINE TO BE PLACED AT 1728MM FROM SRJ AS PER VOSSLOH DRAWING	JAYA S.	VARUN C.	26-OCT-2015
	NOTE ADDED FOR ESP & MARKER BOARDS			
	PLACED DOUBLE HEADS 2.5m APART AS PER PRODUCT OWNER(POLAND) RECOMMENDATION			
	UPDATED CHAINAGE OF S08302 & S08306			
	PLACED 6C BOARD AT 0.5m FROM PLATFORM EDGE AT TEST TRACK AS IN MAINLINE			
	ADDED VIRTUAL SIGNAL CHAINAGES			
	UPDATED NORMING POINT LOCATIONS			
	ADDED SRJ CHAINAGE OF P0818 AND ADJUSTED POINT POSITION			
4.3	ADDED WNRA & CCTV AS PER -100242305_RF_SCH_R89_REV 5.0.	JAYA S.	VARUN C.	14-AUG-2015
	UNMARKED P0823 AS SELF NORMALIZING POINT			
4.2	ADDED PSEUDO PLATFORM AT STABLING LINES.	JAYA S.	VARUN C.	26-MAR-2015
	ADDED NOTE FOR SELF - NORMALIZED POINTS			
	CHANGED REGION 9 (TEST TRACK) "PLATFORM" TO "PSEUDO PLATFORM"			
4.1	ADDED THREE AX HEADS TO MITIGATE ACE COMMUNICATION	JAYA S.	VARUN C.	16-MAR-2015
	UPDATED SYMBOL FOR NEUTRAL SECTION			
	ALIGNED REGION BOUNDARY FOR REFION 8 AND REGION 9			
4.0	RELEASED	RAJNEESH M.	SANTOSH N.	22-JAN-2015
3.1	PESUDO PLATFORM CHANGED TO REAL PLATFORM	RAJNEESH M.	SANTOSH N.	21-JAN-2015
	SIGNAL "S08379" SHIFTED, TO ALIGN WITH SIGNAL "S08375"			
	UPDATED AS PER CUSTOMER COMMENTS NO.DMRC.S&T.CS03.Ph-III.BT-SUB.II.622 DTD 23-DEC-14			
3.0	RELEASED	RAJNEESH M.	JAYA S.	30-OCT-2014
2.3	UPDATED AS PER CUSTOMER COMMENTS DATED 09-OCT-2014	RAJNEESH M.	JAYA S.	27-OCT-2014
	ADDED AXLE COUNTER HEAD AT ENTRY AND EXIT LINE TOWARD DEPOT BOUNDARY TO MITIGATE ACE COMMUNICATION.			
2.2	UPDATED AS PER MOR 100237335-Rev 0.5 DATED 25-09-2014	RAJNEESH M.	SANTOSH N.	26-SEPT-2014
2.1	UPDATED AS PER MOM DATED 22-09-2014	RAJNEESH M.	SANTOSH N.	25-SEPT-2014
2.0	RELEASED	RAJNEESH M.	JAYA S.	01-SEPT-2014
1.4	UPDATED AS PER MOR 100237335 DATED 28-AUG-2014	RAJNEESH M.	JAYA S.	29-AUG-2014
	UPDATED AS PER EQUIPMENT NUMBERING SCHEME 100222667, Rev7.0			
1.3	UPDATED AS PER DMRC COMMENTS DATED 07-AUG-2014	RAJNEESH M.	JAYA S.	26-AUG-2014
1.2	CHANGED REVISION FROM 2.0 TO 1.2	RAJNEESH M.	ARNUT S.	14-JUL-2014
1.1	UPDATED AS PER REVIEWER'S COMMENTS	RAJNEESH M.	SANTOSH N.	20-JUN-2014
	"RM" BORAD REMOVED AT THE ENTRY OF DEPOT			
	VIRTUAL SIGNAL NOMNECLATURE "S" CHANGED TO "VS"			
	FOULING AXLE HEAD "T08016/T08018" REMOVED			
1.0	RELEASED	RAJNEESH M.	ARNUT S.	13-JUN-2014
0.6	-UPDATED AS PER REVIEWERS COMMENT	RAJNEESH M.	ARNUT S.	10-JUN-2014
0.5	-UPDATED AS PER STANDARD LAYOUT & REVISED GAD DTD 28-04-2014	RAJNEESH M.	ARNUT S.	04-JUN-2014
0.4	-UPDATED AS PER LATEST EQUIPMENT NUMBERING SCHEMATIC REV.5.0	RAJNEESH M.	SANTOSH N.	07-MAY-2014
0.3	-UPDATED AS PER DMRC COMMENTS	RAJNEESH M.	DHRUVESH B.	29-APR-2014
0.2	-UPDATED AS PER REVIEWERS COMMENT	RAJNEESH M.	DHRUVESH B.	18-APR-2014
0.1	-FIRST DRAFT	RAJNEESH M.	DHRUVESH B.	16-APR-2014

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100235879	A PRELIMINARIES AND GENERAL REQUIREMENTS	N/S	3	4	6.0

SHEET NO.	DOCUMENT NUMBER	REV.	LIST OF REFERENCE DOCUMENT
06	CBTC MAP SITE SUREVY VERIFICATION REPORT_R9.DOCX	R2.0	SITE SURVEY FOR REGION 9 CONTAINING THE INSTALLED CHAINAGES OF WAYSIDE OBJECTS
06	CBTC MAP SITE SUREVY VERIFICATION REPORT_R8.DOCX	R6.0	SITE SURVEY FOR REGION 8 CONTAINING THE INSTALLED CHAINAGES OF WAYSIDE OBJECTS
05	EMAIL FROM PRAPAN REGARDING CCTV08714 CHAINAGE DATED 10.05.2016	R6.0	RF SCHEMATIC REGION 8-9 (MUKUNDPUR DEPOT)
04	100242305	R6.0	RF SCHEMATIC REGION 8-9 (MUKUNDPUR DEPOT)
03	DRAFT_NORMING POINT_TRACK LAYOUT RECVD. BY MAP TEAM DATED 20/10/2015	-	
02	DEPOT TRACKLAYOUT PLAN DATED 24-04-2014	-	REVISED TRACK LAYOUT FOR MUKUNDPUR DEPOT WITH UTO OPERATION
01	CC-14/VD/ALIGN/SD-104-SHEET02	R2	GENERAL ARRANGEMENT DRAWING HORIZONTAL & VERTICAL ALIGNMENT OF EXIT LINE

REFERENCE DOCUMENT

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SYMBOLS USED IN THE TRACK AND SIGNALLING PLAN

SYMBOL	DESCRIPTION
1 SXXYYY M.D.S.T FOR ELEVATED SXXYYY SXXYYY M.D.S.T FOR UNDERGROUND (WALL MOUNT) SXXYYY	ROUTE INDICATOR — M FOR MAIN LINE GREEN ASPECT — D FOR FIRST DIVERSION VIOLET ASPECT — S FOR SECOND DIVERSION RED ASPECT — T FOR THIRD DIVERSION 3 – ASPECT SIGNAL "XX" = REGION NO. "YYY" = OBJECT NUMBERING (201–300)
2 SXXXXY SAABBB M.D.S.T	POINT PROTECTION SIGNAL "XXXX" = POINT NO. "Y" = DIRECTION OF TRAVEL "AA" = REGION NO. "BBB" = OBJECT NUMBERING (201–300)
3 SXXYYY	SHUNT SIGNAL "XX" = REGION NO. "YYY" = OBJECT NUMBERING (301–500)
4 SXXYYY	DEPENDENT SHUNT SIGNAL "XX" = REGION NO. "YYY" = OBJECT NUMBERING (301–500)
5 SXXYYY FOR ELEVATED/GROUND SXXYYY FOR UNDERGROUND (WALL MOUNT)	BUFFER STOP SIGNAL "XX" = REGION NO. "YYY" = OBJECT NUMBERING (301–350) FOR MAINLINE "YYY" = OBJECT NUMBERING (501–600) FOR DEPOT
6 VSXXYYY	VIRTUAL SIGNAL "XX" = REGION NO. "YYY" = OBJECT NUMBERING (351–400) FOR MAINLINE "YYY" = OBJECT NUMBERING (601–700) FOR DEPOT
7 PXXYY	POINT MACHINE "XX" = REGION NO. "YY" = OBJECT NUMBERING (01–99) = EEA–5 (MAIN LINE) = RDL2 (DEPOT)
8	FOULING POINT (POSITION SHALL BE CONFIRMED AT SITE)
9	ELECTRICAL KEY TRANSMITTER "X" = EKT GROUP NUMBER (I.E. EKT#1 FOR GROUP NO.1) NOTE: STARTING NUMBER "1" IN EVERY STATION
10 TXXYYY	AXLE COUNTER SECTION "XX" = REGION NO. "YYY" = OBJECT NUMBERING (001–200) FOR MAINLINE "YYY" = OBJECT NUMBERING (001–300) FOR DEPOT
11 TXXYYY/TXXYYY	AXLE COUNTER AXLE COUNTER LOCATED WITHIN FOULING AREA

SYMBOL	DESCRIPTION
12	NEUTRAL SECTION
13	BUFFER STOP AND END OF TRACK
14 85 km/h CIVIL SPEED	FROM CIVIL ENGINEERING POINT OF VIEW THE LIMITATION SPEED IS 85 km/h
15 1.500% L=100m	GRADIENT 1.5 GRADIENT LENGTH IS 100 (m)
16 R= 400.000 m L= 800.000 m	CURVE RADIUS IS 400 m CURVE LENGTH IS 800 m
17 KM. 9+911	MILEAGE IS 9911 (m)
18 ITI & OCB	ITI : INTERLOCKING TO INTERLOCKING BORDER (REGION BORDER) & OBJECT CONTROLLER TO OBJECT CONTROLLER BOUNDARY
19 DOWN UP	NORMAL DIRECTION OF TRAFFIC
20	STATION WITH CBI AND OC
21	STATION WITH OC
22 ESPXXYYY	EMERGENCY STOP PLUNGER (ESP) "XX" = REGION NO. "YYY" = OBJECT NUMBERING (501–600) FOR MAINLINE "YYY" = OBJECT NUMBERING (701–750) FOR DEPOT
23 PLATFORM PFXXYYY	PLATFORM "XX" = REGION NO. "YYY" = OBJECT NUMBERING (601–650) FOR MAINLINE "YYY" = OBJECT NUMBERING (751–800) FOR DEPOT
24 PSEUDO PPXXYYY	PSEUDO PLATFORM "XX" = REGION NO. "YYY" = OBJECT NUMBERING (651–700) FOR MAINLINE "YYY" = OBJECT NUMBERING (801–850) FOR DEPOT

SYMBOL	DESCRIPTION
25 XXYYY	NORMING POINT "XX" = REGION No. "YYY" = OBJECT NUMBERING (001–999)
26 WNRAXXYYY	PAIR OF WAYSIDE NETWORK RADIO ASSEMBLY (WNRA) "XX" = REGION NO. "YYY" = OBJECT NUMBERING (001–200) FOR MAINLINE "YYY" = OBJECT NUMBERING (701–900) FOR DEPOT
27 CCTVXXYYY	PAIR OF CLOSED CIRCUIT TELEVISION (CCTV) "XX" = REGION NO. "YYY" = OBJECT NUMBERING (001–200) FOR MAINLINE "YYY" = OBJECT NUMBERING (701–900) FOR DEPOT
28	SINGLE OF LINE–OF–SIGHT (LOS) ANTENNA
29	PAIR OF LINE–OF–SIGHT (LOS) ANTENNA
30	LEAKY FEEDER CABLE
31 AE	ATC ENDS MARKER BOARD
32 AB	ATC BEGINS MARKER BOARD
33 XC	NORMAL STOPPING MARKER BOARD XC = NUMBER OF CAR
34 SR XC FOR MAINLINE	SR = STOPPING REVERSE MARKER BOARD XC = NUMBER OF CAR
35 STOP	STOP BOARD
36	TRAP POINT
37 OCB	OCB : OBJECT CONTROLLER TO OBJECT CONTROLLER (BOUNDARY)
38 PSDXXYYY FOR MAINLINE PSDXXYYY FOR DEPOT	PLATFORM SCREEN DOOR "XX" = REGION NO. "YYY" = OBJECT NUMBERING (601–650) FOR MAINLINE "YYY" = OBJECT NUMBERING (801–850) FOR DEPOT

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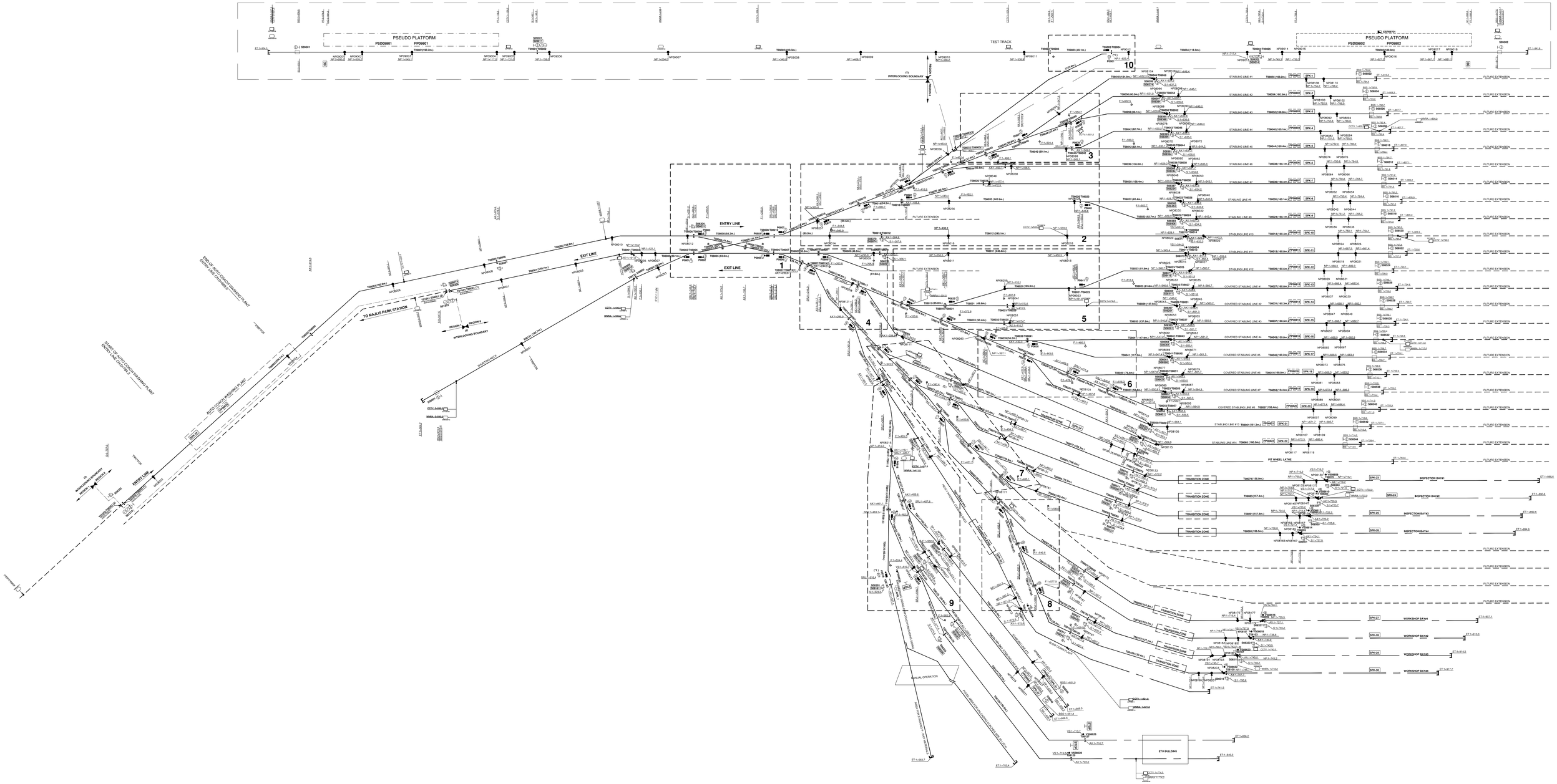

NOTE: 1. POINTS MARKED WITH (?) SHALL SELF NORMALIZE .
2. ESP SHALL BE PLACED AT DCC ROOM.

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CONTENTS: PRELIMINARY AND GENERAL REQUIREMENTS